RESEARCH REPORT · ITERATION 3 · 28 OCT 2026

CV EL research iteration 3

Across eight public skin-lesion datasets, patient details (age, sex, body site and, where recorded, symptoms) add less to a diagnosis once the image model is stronger. The clearest new evidence is a dose–response: the more of HAM10000 a fine-tuned ResNet-18 learns from, the less the patient details add, falling from 9.7 to 0.35 $\times 10^{-3}$ nats. Two predictions failed and are reported as failures: one about imaging modality and one about which metadata are worth collecting.

Bottom line

- **The main effect replicates.** Pre-registered test H1 (a colour-histogram baseline gains more from metadata than ConvNeXt-T) holds on all **6 of 6** independent replication datasets after Holm correction (7 of 7 image sets; the two MILK10k sets show the same lesions).
- **Within deep models it's weaker.** H2 (ResNet-18 gains more than ConvNeXt-T) holds on **3 of 6** independent datasets (4 of 7 image sets).
- **Image quality, not model size, drives it.** More training data and fine-tuning reduce metadata value. Larger frozen models of the same family didn't improve the image model and didn't reduce metadata value.
- **The amount retained differs a lot between datasets.** At ConvNeXt-T, on complete cases, it ranges from about 6% (HIBA) to about 60% (MSK2020) of what metadata alone can do. Clinical photos did *not* retain more than dermoscopy, contrary to the pre-registered prediction.
- **Data hygiene matters.** In PAD-UFES-20, missing values encode the label; removing them cuts metadata-only value by 63%. All main claims are rechecked on complete cases.

Data

Seven independent datasets, plus a clinical-photo version of MILK10k (the same lesions as MILK10k dermoscopic, so not independent). All are public; ADNI is not used. Every split keeps images of the same lesion or patient together. "Complete cases" are images with every metadata variable recorded.

DATASET	IMAGES	COMPLETE CASES	LABEL (POSITIVE RATE)	IMAGING	METADATA
HAM10000	10,015	9,761	melanoma (11%)	dermoscopy	age, sex, site
ISIC 2020	33,126	32,531	melanoma (1.8%)	dermoscopy	age, sex, site

DATASET	IMAGES	COMPLETE CASES	LABEL (POSITIVE RATE)	IMAGING	METADATA
ISIC 2019, HAM images removed	15,316	14,156	melanoma (22%)	dermoscopy	age, sex, site
PAD-UFES-20	2,298	1,089	malignant (47%)	smartphone	age, sex, site, skin type, 6 symptoms
HIBA	1,224	1,085	malignant (44%)	dermoscopy	age, sex, site, skin type
MSK2020	1,174	1,157	malignant (13%)	mixed	age, sex, site
MILK10k dermoscopic	5,117	5,086	malignant (71%)	dermoscopy	age, sex, site
MILK10k clinical	5,117	5,086	malignant (71%)	clinical close-up	age, sex, site

A check by ISIC image ID found no image shared between any two datasets. MSK2020 has no lesion ID, so it is split per image; this is a limitation.

1. Pre-registered tests

Corrected 28 Oct: an earlier version counted the two MILK10k image sets as independent datasets and missed ISIC 2019 among the complete-case H2 failures.

Each test compares the metadata's *added value*: how much adding metadata to an image model's score reduces log-loss on held-out images (in nats; bigger means more useful). The hypotheses were written down before any model was fitted. The first wave was corrected over 6 tests and the second over 8, as registered.

DATASET	H1: HISTOGRAM - CONVNEXT	H2: RESNET-18 - CONVNEXT
ISIC 2020	**2.6 [1.8, 3.4], holds**	0.2 [−0.0, 0.5], p = 0.09
ISIC 2019	**10.2 [5.7, 14.4], holds**	0.6 [−1.1, 2.4], p = 0.24
PAD-UFES-20	**180 [157, 200], holds**	**27.7 [10.3, 45.6], holds**

DATASET	H1: HISTOGRAM - CONVNEXT	H2: RESNET-18 - CONVNEXT
HIBA	**69.6 [39.2, 105.5], holds**	**16.3 [7.5, 25.6], holds**
MSK2020	**13.4 [6.0, 21.4], holds**	3.5 [−1.6, 8.5], p = 0.09
MILK10k dermoscopic	**43.6 [36.9, 50.2], holds**	**6.1 [3.1, 9.2], holds**
MILK10k clinical	**52.9 [45.3, 60.2], holds**	**3.1 [1.2, 4.9], holds**

Differences are $\times 10^{-3}$ nats with 95% grouped-bootstrap intervals; "holds" means significant after Holm correction. HAM10000 was the original pre-registered study (iteration 2) and is not in this table. On complete cases (uncorrected) H1 holds on all 8 image sets, and H2 fails on ISIC 2019, PAD-UFES-20 and MSK2020.

FAILED

P1: clinical photos don't retain more metadata value than dermoscopy

The prediction was that on the same MILK10k lesions, ConvNeXt-T would retain a larger share of metadata value from clinical photos (less lesion detail) than from dermoscopy. It went the other way: the retained fraction was 0.069 for clinical and 0.105 for dermoscopic, a difference of −0.036 [−0.071, −0.004]. Imaging modality does not explain the differences between datasets.

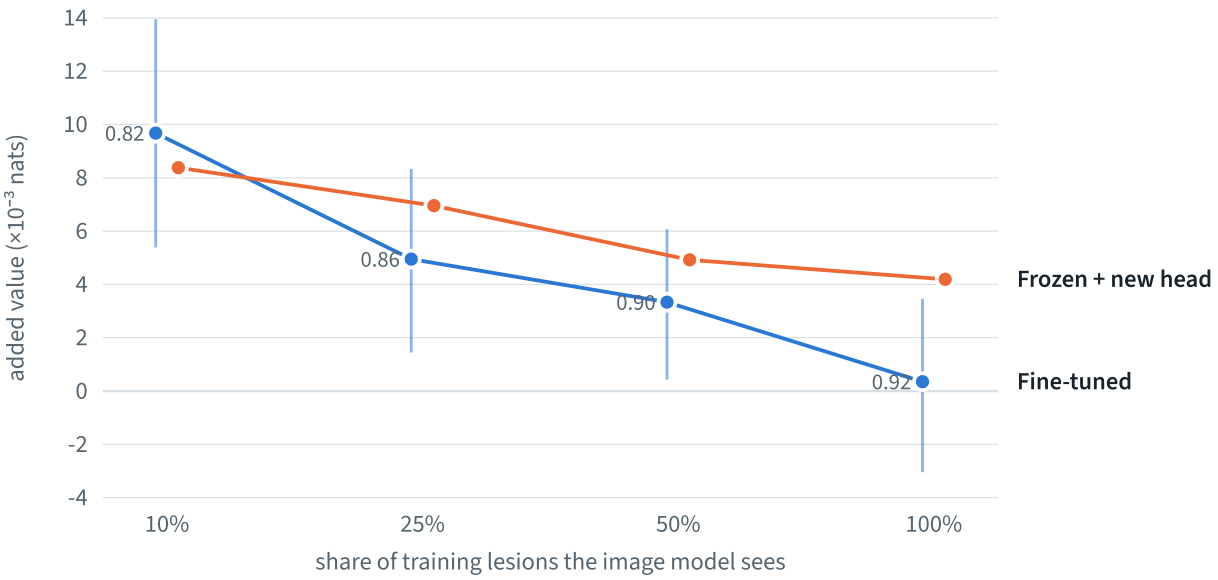
SUPPORTED

P2: measured lesion size (MSK2020) is largely absorbed by the image model

Lesion diameter adds 8.6 [1.3, 16.4] $\times 10^{-3}$ nats to the histogram model but only 1.8 [−1.7, 5.1] to ConvNeXt-T, which retains 21% (the threshold was < 50%). This was registered as exploratory because n is small.

2. Image quality drives the effect, not model size

Metadata added value on HAM10000 (ResNet-18) as the image model sees more data



Blue: fine-tuned, with 95% intervals and image AUROC beside each point. Orange: frozen ImageNet features with a new head, mean of 2 subsamples. The 10–50% points use complete cases (9,761 images); the fine-tuned 100% point is E14 on all 10,015 images. Sources: E31c, E14, E31b.

EXPERIMENT	WHAT CHANGED	RESULT
E31c	Fine-tuned on 10 → 25 → 50 → 100% of HAM10000	**Added value 9.7 → 5.0 → 3.3 → 0.35 as image AUROC rises 0.82 → 0.92**
E14	Frozen vs fine-tuned, HAM10000	ResNet-18: 4.1 → 0.35, drop 3.8 [1.7, 5.9]. ConvNeXt-T: 1.7 → 0.8, drop 0.9 [−1.1, 2.8], not detectable
E14b	Frozen vs fine-tuned, HIBA	Drop significant for both models; ResNet-18 18.9 → 1.4
E14b	Frozen vs fine-tuned, PAD-UFES-20	Drop significant for both, but most value remains (ConvNeXt-T 206 → 124). Run on all images, so missingness leakage is included
E31b	Frozen features, head trained on 10–100% of the data	Same direction, weaker: HAM ResNet-18 8.4 → 4.2; ISIC 2019 noisier

EXPERIMENT	WHAT CHANGED	RESULT
E31a	Bigger frozen models in one family (ResNet-18 → 101; ConvNeXt-T → B)	**No detectable change in any dataset; the image model didn't improve either**
E18	ViT-L/16 (304M parameters) vs ConvNeXt-T, frozen	Image AUROC up by ≤ 0.02; added value slightly lower on ISIC 2020 and ISIC 2019, unchanged elsewhere
E19	Image resolution 16 → 224 px	**Weak manipulation: HAM AUROC is already 0.88 at 16 px. The prediction failed on HAM and ISIC 2019**

In plain terms: whatever makes the image model genuinely better at the task (more data, fine-tuning) removes the metadata's contribution. Adding parameters to frozen features does neither, so "bigger model" is the wrong description and the paper will say "stronger image representation".

3. Controls and data hygiene

> LEAKAGE **PAD-UFES-20 missingness encodes the label**
>
> All 804 images missing sex or skin type are benign (they come from non-biopsied lesions), and "unknown" symptom answers are 88% malignant. On complete cases, metadata-only value falls from 0.416 to 0.155 nats. Smaller missingness effects exist in the other datasets. Every claim is now reported on complete cases as well; PAD results that use all images (E14b, E18, E19) are marked.

- **Shuffled metadata (E32).** Real metadata beat all 50 shuffles on every one of the 8 datasets ($p = 0.02$, the smallest possible with 50 shuffles). Shuffling also raised the rate of implausible records in PAD-UFES-20 from 0.2% to 2.3%, so the shuffle produces the kind of nonsense records it should.
- **Image source as a shortcut.** The HAM10000 result holds when the image source is controlled (iteration 2, E10b).
- **Inputs that give away the label.** The diagnosis-method and biopsy fields are excluded everywhere.

4. Exploratory analyses

These were registered before implementation (REGISTRY_E30_E34.md) but not pre-registered as confirmatory. They generate hypotheses rather than test them.

PROMISING **E30: variables the image can already "see" keep less of their value**

Across 95 dataset × model × variable combinations, the better a variable can be predicted from the image (within each diagnosis class), the smaller the share of its value that survives. Spearman correlation is −0.75, and −0.59 [−0.72, −0.40] after adjusting for the variable's own value and the image AUROC. It holds within every model, from −0.41 to −0.75, so it isn't just restating that stronger models keep less.

Caveats: 65 of 160 rows were excluded because metadata alone had no significant value there, so a retained share is undefined. That rule was fixed in the code before the run but not written into the registry. The retained share is a ratio with a noisy denominator.

FAILED **E34: no useful rule for which metadata to collect**

The fitted rule predicted a variable's added value *worse* than always predicting zero (mean error 0.0101 vs 0.0092 nats, leave-one-dataset-out). The simple baseline "metadata-only value × typical retained share" did best (0.0082; rank correlation 0.74). With seven independent datasets there is little to learn a rule from.

MIXED **E33: across datasets, metadata can hurt**

In 26 train-on-one, test-on-another pairs (age, sex and site harmonised), metadata helped significantly in 14, hurt in 7 and was unclear in 5, ranging from −0.147 to +0.063 nats. The largest harms come from training on MSK2020 (13% malignant) and testing on MILK10k (71%), so they mostly reflect differences in prevalence and label definition rather than a property of the metadata. Within HAM10000, holding out one image source at a time, metadata still helped for 3 of 4 sources.

- **E16, PAD-UFES-20 symptoms.** Symptoms such as bleeding and elevation are partly visible in the image even within a diagnosis class (ConvNeXt within-class AUROC 0.64–0.76). All three registered mechanism hypotheses for PAD's high floor failed.

5. Interpretability (E15)

For a linear head on pooled features, the class activation map (CAM) reproduces the logit exactly; this was checked numerically for each model (r = 1.000). On 2,222 HAM10000 images with expert lesion masks:

- Melanoma evidence is concentrated on the lesion in 84–91% of images, across ResNet-18, ResNet-50 and ConvNeXt-T.
- Under the exact circular-shift null, the overlap is significant for 36–56% of images. The usual pixel-shuffle test claims 82–90%; it is anti-conservative because it ignores spatial smoothness.

- Missed melanomas are smaller: lesion area is 0.33–0.37 of the image vs 0.49–0.52 for detected ones ($p \leq 5\times10^{-12}$).

What can and can't be claimed

CLAIM	STATUS
Metadata add less to strong image representations than to weak ones	**Supported: 6/6 independent replication datasets plus HAM10000, pre-registered, also on complete cases**
Within deep frozen models, metadata value keeps falling	Partly: 3/6 independent datasets
More data and fine-tuning reduce metadata value (dose–response)	**Supported on HAM10000 and HIBA; partial on PAD-UFES-20**
Larger frozen models reduce metadata value	**Not supported**
Imaging modality sets how much value is retained	**Not supported (P1 failed)**
Image-recoverability predicts what is retained	Exploratory evidence only (E30)
A rule can say which metadata to collect	**Not supported (E34)**
Metadata helps under dataset shift	Mixed; confounded with prevalence and label shift
Clinical benefit, or causal effects of metadata	**Not claimed; not tested**

Overnight run, 27–28 Oct

- **Completed: 36 jobs,** 00:09–04:01. Every job wrote its results to a JSON file and finished with exit code 0.
- **Failed: none.**
- **Skipped: 1.** Fine-tuning ResNet-18 on ISIC 2019: its 120-minute estimate would have run past the 06:00 stop.
- The runtime estimates were far too generous; most jobs took a fraction of their budget. Future queues can schedule more.

Paper status and next steps

The IEEE paper (`paper/main.tex` , 6 pages, compiles cleanly) contains the iteration-2 and wave-1 results. The overnight results are **not yet in it**.

1. Add the second-wave tests, the fine-tuning dose–response (as a figure) and the failed P1 to the paper; regenerate all numbers from the JSON results.
2. Reframe from "capacity" to "strength of the image representation", since model size alone did nothing.
3. Record E30–E34 as exploratory in the experiment registry and the novelty ledger.
4. Run the skipped ISIC 2019 fine-tuning, and fine-tuning on MILK10k, to test the dose–response beyond HAM10000.
5. Run another strict reviewer round. The author block still needs your names.

All numbers come from `research/results/*.json` . Status of each run: `research/overnight/STATUS.md` . Auto-generated summary: `research/iteration4/SUMMARY.md` . Pre-registrations: `research/iteration3/` . Exploratory registry: `research/iteration4/REGISTRY_E30_E34.md` .